

Formula Quick Reference

สูตรอ้างอิงฉบับย่อ — คณิตศาสตร์สำหรับ Options

ครอบคลุมบทที่ 1-16 | ใช้เป็น cheat sheet ขณะอ่าน

ส่วนที่ I — เครื่องมือพื้นฐาน (บท 1-3)

% Change และ Asymmetry

% Change

$$\% \Delta = \frac{\text{ใหม่} - \text{เก่า}}{\text{เก่า}} \times 100$$

Recovery needed after $-x\%$

$$r = \frac{x}{100 - x} \times 100\%$$

e.g. ลด 50% → ต้องขึ้น 100%

Logarithm และ e

$$\ln(ab) = \ln a + \ln b \quad \ln(a^n) = n \ln a$$

Log Return

$$r = \ln\left(\frac{S_1}{S_0}\right) = \ln S_1 - \ln S_0$$

บวกข้ามเวลาได้, สมมาตร, เป็น Normal ได้

Options Payoff

$$\text{Long Call} = \max(0, S - K) - C$$

$$\text{Long Put} = \max(0, K - S) - P$$

Put-Call Parity

$$C + Ke^{-rT} = P + S$$

Present Value / Future Value

$$PV = FV \cdot e^{-rT} \quad FV = PV \cdot e^{rT}$$

PV ใน Put-Call Parity

$$PV(K) = Ke^{-rT}$$

ส่วนที่ II — สถิติและความน่าจะเป็น (บท 4–6)

Variance และ Volatility

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n} \quad \sigma = \sqrt{\sigma^2}$$

Annualized Vol

$$\sigma_{\text{annual}} = \sigma_{\text{daily}} \times \sqrt{252}$$

Covariance / Correlation

$$\text{Cov}(X, Y) = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{n}$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} \in [-1, +1]$$

Expected Value

$$E[X] = \sum_i x_i P(x_i)$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Probability

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Conditional / Bayes

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

Combinations

$$C(n, k) = \binom{n}{k} = \frac{n!}{k! (n - k)!}$$

Distributions

Binomial

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Normal PDF

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Z-Score / CDF

$$Z = \frac{x - \mu}{\sigma} \quad N(d) = P(Z \leq d)$$

Lognormal

$$\ln(S) \sim \mathcal{N}(\mu, \sigma^2) \Rightarrow S \sim \text{Lognormal}$$

ส่วนที่ III — แคลคูลัส (บท 7)

Derivatives

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^{ax}) = ae^{ax}$$

$$\frac{d}{dx}[\ln(x)] = \frac{1}{x}$$

$$\frac{d}{dx}[f \cdot g] = f'g + fg'$$

Integration

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int e^x dx = e^x + C$$

Option price as integral

$$C = e^{-rT} \int_0^\infty \max(0, S - K) \cdot f(S) dS$$

Greeks (Partial Derivatives of C)

Greek	Formula	Meaning
Δ	$\partial C / \partial S = N(d_1)$	Rate vs S
Γ	$\partial^2 C / \partial S^2$	Curvature
Θ	$-\partial C / \partial T$	Time decay
ν	$\partial C / \partial \sigma$	Vol sensitivity
ρ	$\partial C / \partial r$	Rate sensitivity

Taylor Expansion (P&L Attribution)

$$\Delta V \approx \Delta \cdot \Delta S + \frac{1}{2} \Gamma (\Delta S)^2 + \Theta \cdot \Delta t + \nu \cdot \Delta \sigma$$

Directional + Convexity + Time Decay + Vol P&L

ส่วนที่ IV — Linear Algebra + Regression (บท 8–9)

Portfolio Variance

$$\sigma_p^2 = \mathbf{w}^\top \Sigma \mathbf{w}$$

$\mathbf{w}$ = weight vector, Σ = covariance matrix

2-Asset

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \text{Cov}_{12}$$

Matrix Inverse (2x2)

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

OLS Regression

$$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Matrix OLS

$$\hat{\beta} = (X^\top X)^{-1} X^\top \mathbf{y}$$

t-statistic

$$t = \frac{\hat{\beta}}{\text{se}(\hat{\beta})} \quad |t| > 2 \Rightarrow \text{significant}$$

R^2 (Goodness of Fit)

$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}} \in [0, 1]$$

ส่วนที่ V — Optimization (บท 10–11)

Linear Programming

$$\max \mathbf{c}^\top \mathbf{x} \text{ s.t. } A\mathbf{x} \leq \mathbf{b}, \mathbf{x} \geq 0$$

Optimal always at a corner point of feasible region

Gradient Descent

$$\mathbf{x}_{n+1} = \mathbf{x}_n - \alpha \nabla f(\mathbf{x}_n)$$

α = learning rate, converges for convex f

Lagrange Multipliers

$$\mathcal{L}(x, \lambda) = f(x) - \lambda g(x)$$

Markowitz (equality form)

$$\min \mathbf{w}^\top \Sigma \mathbf{w} \text{ s.t. } \mathbf{1}^\top \mathbf{w} = 1, \boldsymbol{\mu}^\top \mathbf{w} = r^*$$

KKT Conditions

1. Stationarity: $\nabla f + \sum \mu_i \nabla g_i = 0$

2. Primal feasibility: $g_i(\mathbf{x}) \leq 0$

3. Dual feasibility: $\mu_i \geq 0$

4. Complementary slackness: $\mu_i \cdot g_i = 0$

ส่วนที่ VI — Time Series + Stochastic Calculus (บท 12–14)

Volatility Models

EWMA

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

$\alpha + \beta < 1 \rightarrow$ mean reversion; long-run $\sigma = \sqrt{\omega/(1-\alpha-\beta)}$

Newton-Raphson IV Solver

$$\sigma_{n+1} = \sigma_n - \frac{C_{BS}(\sigma_n) - C_{mkt}}{\mathcal{V}(\sigma_n)}$$

Vega = $\partial C / \partial \sigma$; converge in 3–5 iterations

Stochastic Processes

GBM (Geometric Brownian Motion)

$$dS = \mu S dt + \sigma S dW_t$$

Ito's Lemma

$$df = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial S} dS + \frac{1}{2} \frac{\partial^2 f}{\partial S^2} (dS)^2$$

$(dW)^2 = dt$ (Ito correction)

Black-Scholes PDE

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

Theta + Gamma P&L + Delta P&L = risk-free return

ส่วนที่ VII — Black-Scholes + Monte Carlo (บท 15–16)

Black-Scholes Call / Put

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} \quad d_2 = d_1 - \sigma\sqrt{T}$$

BS with Dividends (Merton)

$$C = S e^{-qT} N(d_1) - K e^{-rT} N(d_2)$$

replace S with $S e^{-qT}$; d_1 uses $(r-q+\sigma^2/2)$

Greeks (Closed Form)

Greek	Call	Put
Δ	$N(d_1)$	$N(d_1) - 1$
Γ	$N'(d_1)/(S\sigma\sqrt{T})$	
ν	$S\sqrt{T} N'(d_1)$	
Θ	$-\frac{SN'(d_1)\sigma}{2\sqrt{T}} - rKe^{-rT}N(d_2)$	similar $+rKe^{-rT}N(-d_2)$
ρ	$Ke^{-rT}N(d_2)$	$-Ke^{-rT}N(-d_2)$

Monte Carlo Pricing

GBM discrete simulation

$$S_{t+\Delta t} = S_t \exp\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z\right]$$

$Z \sim N(0,1)$ i.i.d.

MC option price

$$\hat{C} = e^{-rT} \frac{1}{N} \sum_{i=1}^N \max(0, S_T^{(i)} - K)$$

Standard Error

$$SE = \frac{\hat{\sigma}_{\text{payoff}}}{\sqrt{N}}$$

4x paths $\rightarrow$ 2x precision ($SE \propto 1/\sqrt{N}$)

Control Variates

$$Y^* = Y_{\text{Asian}} - \hat{\beta}(Y_{\text{Vanilla,MC}} - Y_{\text{Vanilla,BS}})$$